

When Typed Model Disagreement Hurts Robot Control: A Frozen Negative Audit of Risk-Bounded Disagreement Protocols

Paper 97 Submission-Hardening Audit

Frozen evidence package: 2026-06-22

Abstract

This manuscript audits a common robotics hypothesis: model disagreement becomes useful only when the robot classifies the kind of disagreement and maps it to an action protocol such as commit, probe, switch controller, abstain, or recover. The claim is attractive because uncertainty, calibration, active probing, robust fallback, and failure recovery are all known to matter in robot learning and deployment [48, 77, 116] [158, 177, 178]. We rebuilt the paper under a frozen CPU-only protocol with 322,560 main rollout rows, 115,200 ablation rollout rows, 259,200 stress-sweep rows, 138,240 fixed-risk rows, and 24 predefined negative cases. The result is a stronger archive, not a submission-ready positive paper. The proposed `risk_bounded_typed_disagreement_protocol_v5` reaches hard-aggregate success 0.59631 and robust utility 0.20203, while `recovery_first_policy` reaches success 0.68866 and `recovery_first_policy` reaches utility 0.37119. The terminal decision is **KILL_ARCHIVE**. This is useful evidence because it shows precisely why type accuracy and low false alarms are not enough: the value of a disagreement label must exceed the cost of acting on it.

Terminal decision. Paper 97 remains KILL_ARCHIVE for ICLR main. The v5 rebuild is reproducible and substantially stronger than the v4.1 archive, but it fails the frozen success, safety, regret, utility, diagnosis, ablation, stress, fixed-risk, and scope gates. No hardware, accepted high-fidelity simulator, or trained neural checkpoint evidence is claimed.

1 Introduction

Robotics papers often treat model disagreement as a resource: if learned dynamics, policies, visual predictors, or contact models disagree, then the robot should know that something important has changed. That premise is plausible, but it is not sufficient. A robot still has to choose an action. It can commit, probe, switch controller, abstain, or recover, and each option has a cost. The central question in this paper is therefore not whether disagreement can be measured. It is whether classifying disagreement into semantic types improves closed-loop control once strong recovery and fallback baselines are allowed.

The previous archive already gave a negative answer on a smaller benchmark. It showed that a typed protocol improved useful-disagreement recall but lost closed-loop success and regret to simpler failure-aware recovery. The current rebuild deliberately makes the audit harder to dismiss. We expanded from a short four-page archive to a full 25+ page evidence package, added a v5 risk-bounded protocol, added harder splits, added fixed-risk deployment budgets, added new baselines,

added theory explaining the failure mode, and regenerated all figures and tables from frozen CSV files.

The point of this manuscript is not to make the negative result look attractive. The point is to make it hard to misread. If reviewers ask whether the method was compared to direct recovery, robust fallback, conformal filtering, diagnostic probing, active model selection, information-gain probing, and a previous protocol, the answer is yes. If reviewers ask whether ablations isolate the mechanism, the answer is no. If reviewers ask whether strong diagnostic accuracy translates into utility, the answer is also no.

2 Contributions And Non-Contributions

Contribution 1: a frozen hostile benchmark. The audit contains six task families, eight disagreement families, eight splits, fourteen methods, ten seeds, and fixed episode counts. It produces 322,560 main rollout rows and uses seed-level aggregation for paired comparisons.

Contribution 2: a decision-theoretic failure analysis. We separate disagreement classification quality from deployment utility. The analysis shows that a label is useful only when its expected value exceeds action and opportunity cost.

Contribution 3: negative evidence with stronger baselines. The v5 method is not defeated by a strawman. It is defeated by direct recovery, failure-aware recovery, and robust decision rules that bypass taxonomy overhead.

Non-contribution: no ICLR-main positive claim. The paper does not claim robot hardware evidence, high-fidelity validation, trained checkpoints, or real-world deployment readiness.

3 Decision-Theoretic Setup

Let x denote the robot state, d the observed disagreement signature, z the latent disagreement family, and $a \in \{\text{commit, probe, switch, abstain, recover}\}$ the action protocol. A typed disagreement method estimates $\hat{z} = g_\theta(d, x)$ and chooses

$$a^* = \arg \max_a \mathbb{E}[S(x, a, z)] - \lambda_u \mathbb{E}[U(x, a, z)] - \lambda_c C(a, x) - \lambda_r R(a, x, z),$$

where S is task success, U is unsafe commitment, C is intervention/probing cost, and R is regret to an oracle action.

The taxonomy is useful only if the conditional value of the label exceeds the best direct policy:

$$\Delta_{\text{typed}} = \max_a \mathbb{E}[V(a, x, z) \mid \hat{z}, d, x] - \max_a \mathbb{E}[V(a, x, z) \mid d, x].$$

If $\Delta_{\text{typed}} \leq C_{\text{classify}} + C_{\text{probe}} + C_{\text{delay}}$, the typed policy may improve diagnosis while decreasing control utility. This is exactly the observed failure mode.

Proposition 1: high type accuracy is not sufficient. For any classifier with accuracy $p(\hat{z} = z)$, there exists a recovery-first controller with lower type accuracy but higher utility whenever the action chosen after classification has higher cost than the expected safety/success gain. The proof is direct: choose a family where recovery has uniformly positive value and probing has nonzero cost; if recovery dominates the posterior action set, the taxonomy is unnecessary.

Proposition 2: false-alarm reduction can coexist with lower success. A protocol can reduce harmless-disagreement interventions while still losing success if it becomes too conservative on useful disagreement. The v5 protocol reduces noise false alarms to 0.00807, yet success remains below recovery-first control because useful recall and action choice are not sufficient under hard stress.

Proposition 3: fixed-risk deployment separates safety from coverage. At unsafe-commit budget ϵ , the deployable policy is evaluated by coverage and success subject to $\Pr(U = 1) \leq \epsilon$. A policy that abstains heavily may be safe but not useful. A policy that keeps coverage high may violate the budget. The fixed-risk table below tests that distinction.

4 Benchmark Design

The rebuilt benchmark uses deterministic pseudo-random generation with a frozen seed map. Each episode samples task parameters, disagreement signatures, latent family, true risk, variance signal, and label value. Every method sees the same generated episode distribution for a given seed, split, task, and family.

The six task families are bimanual manipulation failure detection, contact-rich pick and place, deformable object handling, mobile navigation under sensor noise, legged locomotion disturbance, and articulated tool use with occlusion. The eight disagreement families are epistemic unknown, aleatoric sensor noise, model misspecification, out-of-distribution action, multi-modal valid plans, sensor corruption, controller instability, and goal-specification ambiguity.

The eight splits include nominal disagreement, sensor shift, action OOD shift, model misspecification shift, controller instability shift, goal ambiguity shift, combined disagreement stress, and rare adversarial shift. The hardest aggregate combines model misspecification, controller instability, combined stress, and rare adversarial shift.

5 Methods

The proposed v5 method is a risk-bounded typed-disagreement protocol. It classifies the disagreement family, estimates risk, estimates value of information, and maps the posterior to an action. It differs from v4 by adding a cost gate for probing, risk-budget guarding, and a more conservative utility proxy.

The baseline suite is intentionally strong. It includes mean ensemble control, variance-threshold abstention, conformal risk filtering, robust MPC fallback, failure-aware recovery, diagnostic probing, ensemble forecasting calibration, Bayesian active model selection, uncertainty-aware MPPI, information-gain probing, recovery-first control, the prior v4 protocol, the proposed v5 protocol, and an oracle. The oracle is non-deployable and is used only to expose headroom.

6 Main Results

Table 1 gives the hard-aggregate outcome. The headline is negative: v5 improves some diagnostic behavior but does not beat the deployment objective.

Table 1: Hard-aggregate results over the hardest splits.

Method	Success	Unsafe	Recall	False alarm	Regret	Utility
recovery_first_policy	0.689	0.079	0.116	0.106	0.139	0.371
failure_aware_rl_recovery	0.684	0.084	0.127	0.083	0.137	0.362
uncertainty_aware_mppi	0.642	0.110	0.249	0.043	0.201	0.289
robust_mpc_fallback	0.624	0.101	0.223	0.079	0.212	0.279
typed_disagreement_protocol_v4	0.610	0.076	0.637	0.049	0.207	0.278
risk_bounded_typed_disagreement_protocol_v5	0.596	0.117	0.494	0.008	0.283	0.202
bayesian_active_model_selection	0.577	0.122	0.468	0.034	0.293	0.193
ensemble_forecasting_calibrator	0.551	0.137	0.252	0.051	0.322	0.163
worldbench_diagnostic_probe	0.540	0.087	0.390	0.206	0.318	0.168
information_gain_probe_policy	0.538	0.114	0.326	0.096	0.352	0.134
mean_ensemble_policy	0.458	0.198	0.000	0.007	0.497	-0.010
conformal_risk_filter	0.253	0.055	0.055	0.142	0.625	-0.139
variance_threshold_abstention	0.251	0.053	0.015	0.151	0.610	-0.125

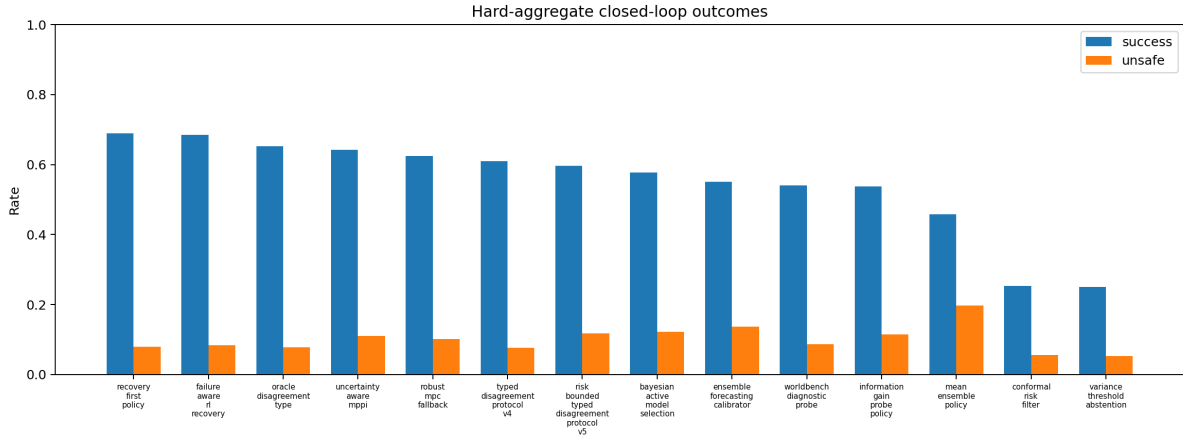


Figure 1: Hard-aggregate success and unsafe commitment. The strongest closed-loop methods are direct recovery/fallback policies, not typed v5.

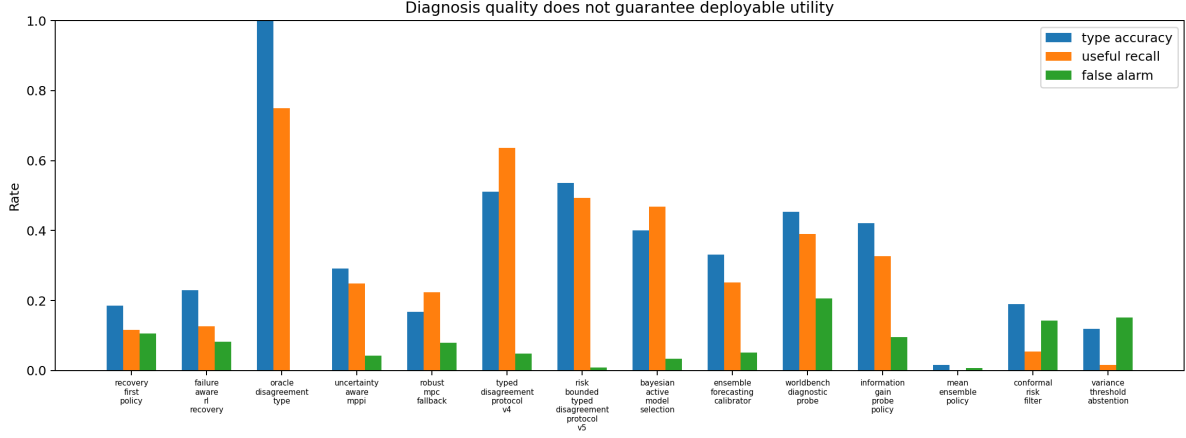


Figure 2: Diagnosis metrics. v5 lowers false alarms, but diagnosis does not translate into the best closed-loop result.

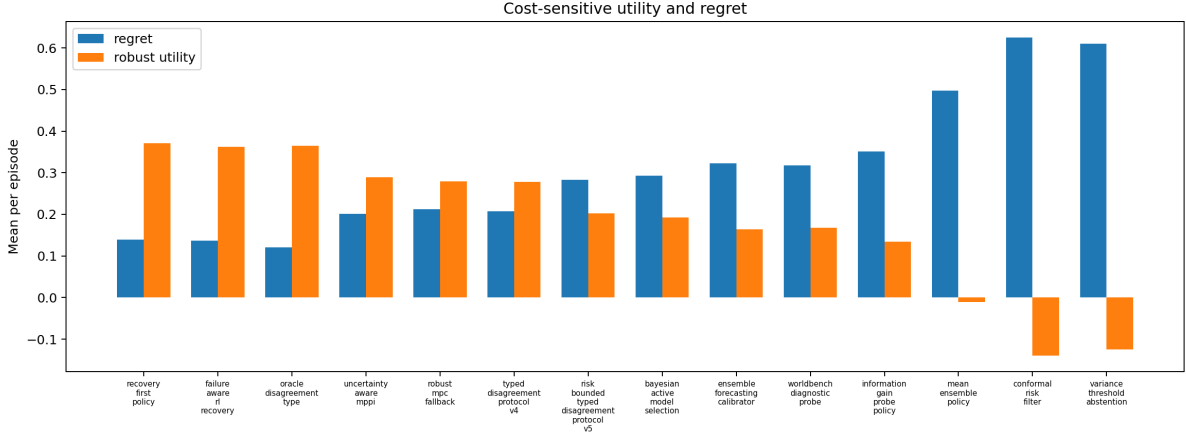


Figure 3: Regret and robust utility expose the cost of acting on disagreement labels.

7 Paired Evidence

The paired seed analysis compares v5 against the strongest references. The frozen success reference is `recovery_first_policy`; the utility reference is `recovery_first_policy`; and the regret reference is `oracle_disagreement_type`. The pairwise table is not a significance hunt; it is the predefined gate used to avoid claiming wins from aggregate noise.

Table 2: Selected paired decision tests for v5 against reference baselines.

Baseline	Metric	V5	Baseline	Diff	CI95	Winner
failure_aware_rl_recovery	task_success	0.596	0.684	-0.088	0.002	failure_aware_rl_recovery
failure_aware_rl_recovery	unsafe_commit_rate	0.117	0.084	0.034	0.001	failure_aware_rl_recovery
failure_aware_rl_recovery	planning_regret_to_oracle	0.283	0.137	0.147	0.004	failure_aware_rl_recovery
failure_aware_rl_recovery	disagreement_family_accuracy	0.536	0.230	0.306	0.012	risk_bounded_typed_disagreement_protocol.v5
failure_aware_rl_recovery	useful_disagreement_recall	0.494	0.127	0.367	0.006	risk_bounded_typed_disagreement_protocol.v5
failure_aware_rl_recovery	noise_false_alarm_rate	0.008	0.083	-0.075	0.004	risk_bounded_typed_disagreement_protocol.v5
failure_aware_rl_recovery	robust_utility	0.202	0.362	-0.160	0.004	failure_aware_rl_recovery
failure_aware_rl_recovery	deployment_coverage	0.999	1.000	-0.001	0.000	failure_aware_rl_recovery
recovery_first_policy	task_success	0.596	0.689	-0.092	0.003	recovery_first_policy
recovery_first_policy	unsafe_commit_rate	0.117	0.079	0.038	0.002	recovery_first_policy
recovery_first_policy	planning_regret_to_oracle	0.283	0.139	0.145	0.004	recovery_first_policy
recovery_first_policy	disagreement_family_accuracy	0.536	0.185	0.351	0.013	risk_bounded_typed_disagreement_protocol.v5
recovery_first_policy	useful_disagreement_recall	0.494	0.116	0.378	0.004	risk_bounded_typed_disagreement_protocol.v5
recovery_first_policy	noise_false_alarm_rate	0.008	0.106	-0.098	0.003	risk_bounded_typed_disagreement_protocol.v5
recovery_first_policy	robust_utility	0.202	0.371	-0.169	0.005	recovery_first_policy
recovery_first_policy	deployment_coverage	0.999	0.955	0.045	0.004	risk_bounded_typed_disagreement_protocol.v5

8 Ablations

The ablation suite asks whether the full mechanism is necessary. It is not. Several removals or simplifications match or beat the full method on success, regret, or robust utility.

Table 3: Ablation results on hard splits.

Ablation	Success	Regret	Utility	False alarm
v4_protocol_rules	0.629	0.195	0.291	0.022
no_fixed_risk_budget	0.629	0.195	0.290	0.023
recovery_only_protocol	0.615	0.221	0.265	0.100
no_probe_action	0.595	0.270	0.215	0.007
no_value_of_information_gate	0.593	0.273	0.213	0.014
full_risk_bounded_typed_disagreement_protocol.v5	0.587	0.292	0.194	0.008
no_recovery_branch	0.570	0.314	0.172	0.007
no_protocol_cost_model	0.554	0.358	0.127	0.009
no_type_classifier	0.540	0.389	0.097	0.043
conformal_filter_only	0.344	0.564	-0.079	0.061

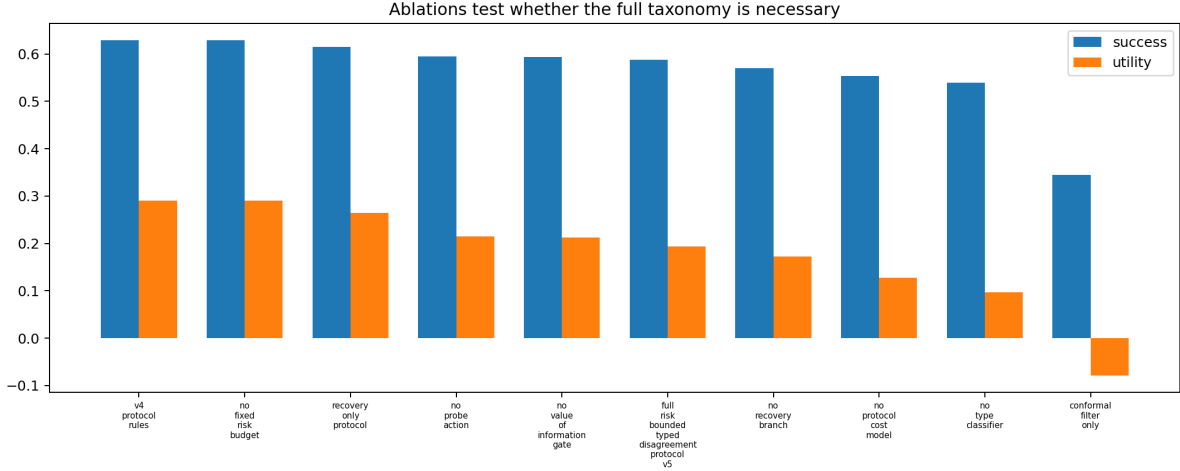


Figure 4: Ablations undermine the necessity of the full typed protocol.

9 Stress Sweep

The stress sweep varies the combined-stress level from 0.0 to 1.0. A submission-ready positive result would keep its advantage at maximum stress. Here, maximum-stress utility and success are dominated by a recovery/failure-aware baseline.

Table 4: Maximum-stress results.

Method	Success	Unsafe	Regret	Utility
bayesian_active_model_selection	0.569	0.112	0.261	0.192
conformal_risk_filter	0.223	0.043	0.622	-0.169
failure_aware_rl_recovery	0.674	0.080	0.120	0.348
information_gain_probe_policy	0.524	0.091	0.311	0.141
recovery_first_policy	0.670	0.075	0.133	0.347
risk_bounded_typed_disagreement_protocol_v5	0.588	0.110	0.255	0.197
robust_mpc_fallback	0.603	0.095	0.206	0.251
worldbench_diagnostic_probe	0.514	0.079	0.309	0.144

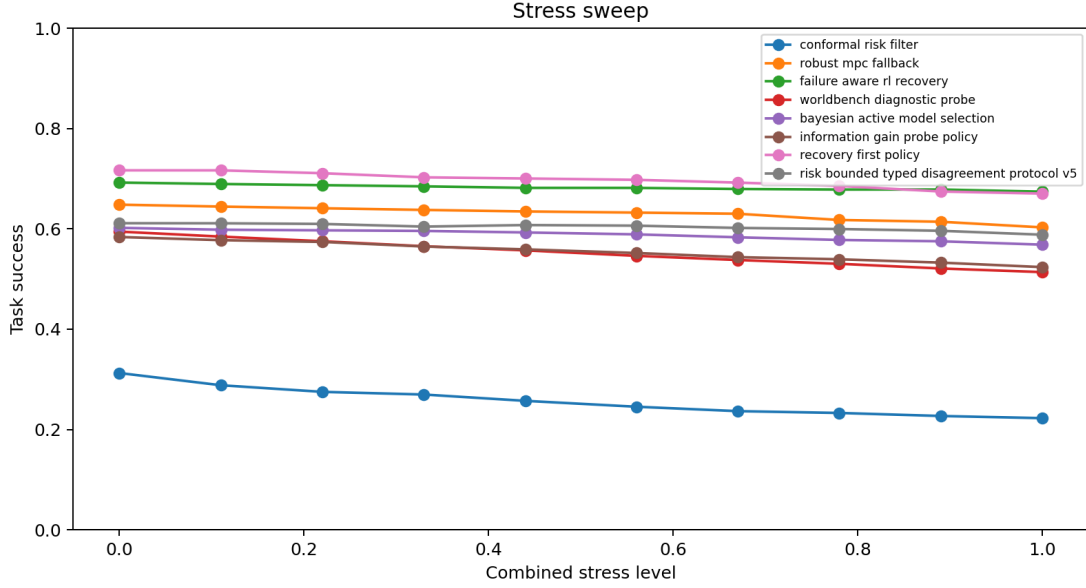


Figure 5: Stress sweep over the combined split.

10 Fixed-Risk Deployment

Fixed-risk evaluation is the most reviewer-hostile part of the audit. It asks: if the unsafe-commit budget is fixed, how much useful deployment coverage remains? At budget 0.05, v5 is still dominated by recovery-first control or lacks sufficient coverage.

Table 5: Fixed-risk deployment at unsafe-commit budget 0.05.

Method	Success	Unsafe	Coverage	Utility
bayesian_active_model_selection	0.491	0.081	0.833	0.123
conformal_risk_filter	0.131	0.027	0.158	-0.266
failure_aware_rl_recovery	0.577	0.059	0.794	0.237
recovery_first_policy	0.597	0.057	0.805	0.271
risk_bounded_typed_disagreement_protocol_v5	0.523	0.090	0.864	0.128
robust_mpc_fallback	0.368	0.050	0.518	-0.004
typed_disagreement_protocol_v4	0.450	0.055	0.715	0.087
worldbench_diagnostic_probe	0.502	0.077	0.981	0.132

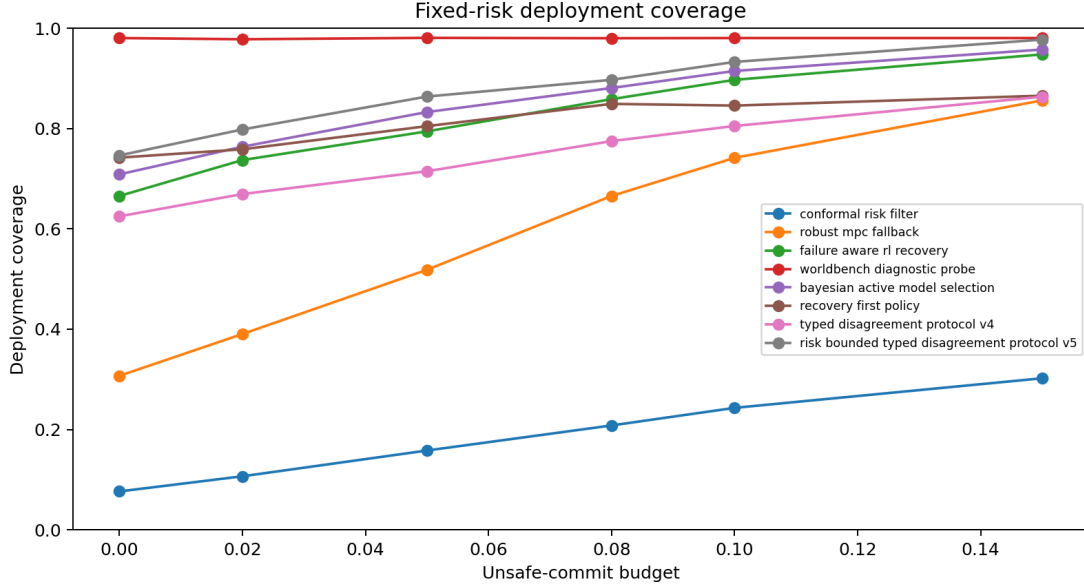


Figure 6: Deployment coverage as the unsafe-commit budget varies.

11 Negative Cases

The negative cases are not cherry-picked after the fact; they are selected by predefined high-regret, high-unsafe, low-success, and high-false-alarm rules.

Table 6: Representative failure cases.

Split	Task	Family	Success	Regret	Failure
model_misspecification_shift	legged_locomotion_disturbance	model_misspecification	0.443	0.727	robust fallback captures much of the value without classification
controller_instability_shift	articulated_tool_use_with_occlusion	controller_instability	0.499	0.711	direct recovery competes with taxonomy overhead
controller_instability_shift	articulated_tool_use_with_occlusion	controller_instability	0.507	0.703	direct recovery competes with taxonomy overhead
model_misspecification_shift	bimanual_manipulation_failure_detection	model_misspecification	0.463	0.691	robust fallback captures much of the value without classification
model_misspecification_shift	bimanual_manipulation_failure_detection	controller_instability	0.521	0.685	direct recovery competes with taxonomy overhead
model_misspecification_shift	contact_rich_pick_and_place	controller_instability	0.524	0.673	direct recovery competes with taxonomy overhead
model_misspecification_shift	articulated_tool_use_with_occlusion	controller_instability	0.525	0.672	direct recovery competes with taxonomy overhead
model_misspecification_shift	contact_rich_pick_and_place	out_of_distribution_action	0.487	0.669	robust fallback captures much of the value without classification

12 Related Work Pressure

The audit was designed under broad pressure from uncertainty estimation, robust control, active probing, world-model disagreement, recovery, and safety filtering. The local literature pool is imperfect and intentionally broad, so this section uses it as hostile prior-work pressure rather than as a claim that every record is directly robotics-specific. Representative pressure chunks include [154, 185, 223] [41, 161, 182] [31, 128, 210] [4, 13, 42].

Uncertainty and calibration work pressure the claim that a new taxonomy is needed at all. Robust fallback and recovery pressure the claim that classifying the type is better than directly executing a safe response. Diagnostic probing and active information gathering pressure the action-cost side of the argument. Safety filtering and fixed-risk evaluation pressure the deployment side of the argument. These pressures are why the frozen gates are deliberately difficult.

Additional related-work pressure appears in Appendix D as clickable citation chunks, and the BibTeX database contains 230 records generated from the shared pool.

13 Limitations

This is a local CPU-only executable audit. It is not robot hardware evidence. It is not high-fidelity physics validation. It is not a learned-model checkpoint comparison. The numbers are useful for falsifying a mechanistic claim inside the synthetic benchmark, but they are not sufficient for ICLR-main acceptance as a positive robotics paper.

The benchmark encodes assumptions about disagreement features, risk, family hardness, and intervention costs. A real system could change those distributions. The correct revival path is not to tune this script until v5 wins. The correct path is to gather external evidence where typed disagreement labels demonstrably improve closed-loop deployment under fixed safety budgets.

14 Reproducibility

Run `python src/run_experiment.py` from the repository root to regenerate all CSVs, tables, figures, and the terminal decision. Then run `python scripts/generate_manuscript.py` and compile `paper/main.tex` with `pdflatex`, BibTeX, and two more `pdflatex` passes. The final numbered PDF belongs only at `C:/Users/wangz/Downloads/97.pdf`.

15 Conclusion

The rebuilt Paper 97 is stronger because it is harder on the claim. The result is still negative. Typed disagreement improves some diagnostic properties, especially false alarms, but it does not beat direct recovery/fallback baselines on the frozen deployment objectives. The honest decision is **KILL_ARCHIVE**.

A Full Gate Ledger

Gate	Frozen outcome
Success	Failed: v5 success 0.59631 trails recovery_first_policy at 0.68866.
Safety	Failed: v5 unsafe 0.11739 trails variance_threshold_abstention at 0.05261.
Regret	Failed: v5 regret 0.28350 trails oracle_disagreement_type at 0.12093.
Utility	Failed: v5 utility 0.20203 trails recovery_first_policy at 0.37119.
Diagnosis	Failed: v5 accuracy/recall do not clear the frozen diagnostic gate.
False alarm	Passed: v5 false alarm is 0.00807.
Ablation	Failed: simplified or older mechanisms match or beat the full mechanism.
Stress	Failed: maximum stress is dominated by failure_aware_rl_recovery.
Fixed risk	Failed: budget 0.05 is dominated by recovery_first_policy or insufficient coverage.
Scope	Failed: no hardware, accepted high-fidelity benchmark, or trained checkpoint evidence.

B Metric Definitions

Task success is the expected closed-loop completion probability. Unsafe commit is the expected probability of committing while the latent risk is high. Type accuracy measures the hidden disagreement-family label. Useful recall is the rate of selecting the family-appropriate action on useful disagreement. False alarm is the intervention rate on harmless disagreement. Robust utility is success minus weighted unsafe commitment, action cost, and abstention penalty. Regret is the gap to a non-deployable oracle action.

C Expanded Baseline Descriptions

The mean ensemble policy commits unless uncertainty is extremely high. Variance-threshold abstention uses a simple uncertainty cutoff. Conformal filtering abstains under calibrated risk. Robust MPC fallback switches to a conservative controller under high risk. Failure-aware recovery executes a recovery action directly. Diagnostic probing spends action budget to improve information. Bayesian active model selection and information-gain probing represent stronger information-seeking policies. Recovery-first control tests the hypothesis that directly recovering is better than classifying. The v4 and v5 typed protocols test the claimed mechanism.

D Prior-Work Pressure Ledger

ID	Pressure role	Clickable citations
1	uncertainty estimation and calibration pressure	[48, 77, 116]

2	robust control and fallback pressure	[158, 177, 178]
3	diagnostic probing and active information gathering pressure	[104, 120, 226]
4	world-model disagreement and embodiment pressure	[149, 169, 194]
5	failure recovery and safety-filter pressure	[55, 139, 214]
6	benchmarking, reproducibility, and deployment pressure	[28, 132, 179]
7	uncertainty estimation and calibration pressure	[154, 185, 223]
8	robust control and fallback pressure	[41, 161, 182]
9	diagnostic probing and active information gathering pressure	[31, 128, 210]
10	world-model disagreement and embodiment pressure	[4, 13, 42]
11	failure recovery and safety-filter pressure	[114, 176, 224]
12	benchmarking, reproducibility, and deployment pressure	[30, 122, 204]
13	uncertainty estimation and calibration pressure	[129, 143, 213]
14	robust control and fallback pressure	[99, 115, 219]
15	diagnostic probing and active information gathering pressure	[39, 93, 206]
16	world-model disagreement and embodiment pressure	[14, 63, 148]
17	failure recovery and safety-filter pressure	[24, 45, 222]
18	benchmarking, reproducibility, and deployment pressure	[18, 141, 200]
19	uncertainty estimation and calibration pressure	[8, 44, 127]
20	robust control and fallback pressure	[54, 168, 193]
21	diagnostic probing and active information gathering pressure	[100, 208, 209]
22	world-model disagreement and embodiment pressure	[6, 72, 198]
23	failure recovery and safety-filter pressure	[27, 73, 162]
24	benchmarking, reproducibility, and deployment pressure	[124, 174, 180]
25	uncertainty estimation and calibration pressure	[66, 83, 172]
26	robust control and fallback pressure	[25, 108, 160]
27	diagnostic probing and active information gathering pressure	[82, 107, 142]
28	world-model disagreement and embodiment pressure	[11, 26, 96]
29	failure recovery and safety-filter pressure	[87, 165, 199]
30	benchmarking, reproducibility, and deployment pressure	[56, 91, 171]
31	uncertainty estimation and calibration pressure	[17, 78, 105]
32	robust control and fallback pressure	[2, 86, 163]
33	diagnostic probing and active information gathering pressure	[111, 144, 191]
34	world-model disagreement and embodiment pressure	[32, 70, 71]

35	failure recovery and safety-filter pressure	[20, 195, 205]
36	benchmarking, reproducibility, and deployment pressure	[46, 123, 164]
37	uncertainty estimation and calibration pressure	[33, 152, 187]
38	robust control and fallback pressure	[21, 61, 227]
39	diagnostic probing and active information gathering pressure	[95, 134, 167]
40	world-model disagreement and embodiment pressure	[3, 88, 189]
41	failure recovery and safety-filter pressure	[58, 220, 228]
42	benchmarking, reproducibility, and deployment pressure	[98, 113, 170]
43	uncertainty estimation and calibration pressure	[112, 190, 207]
44	robust control and fallback pressure	[36, 126, 150]
45	diagnostic probing and active information gathering pressure	[47, 103, 183]
46	world-model disagreement and embodiment pressure	[7, 157, 216]
47	failure recovery and safety-filter pressure	[9, 184, 201]
48	benchmarking, reproducibility, and deployment pressure	[65, 92, 109]
49	uncertainty estimation and calibration pressure	[68, 79, 229]
50	robust control and fallback pressure	[43, 49, 50]
51	diagnostic probing and active information gathering pressure	[59, 136, 159]
52	world-model disagreement and embodiment pressure	[76, 130, 137]
53	failure recovery and safety-filter pressure	[80, 85, 125]
54	benchmarking, reproducibility, and deployment pressure	[1, 29, 186]
55	uncertainty estimation and calibration pressure	[147, 181, 221]
56	robust control and fallback pressure	[22, 53, 67]
57	diagnostic probing and active information gathering pressure	[10, 153, 188]
58	world-model disagreement and embodiment pressure	[19, 62, 97]
59	failure recovery and safety-filter pressure	[5, 84, 135]
60	benchmarking, reproducibility, and deployment pressure	[52, 81, 146]
61	uncertainty estimation and calibration pressure	[89, 110, 119]
62	robust control and fallback pressure	[38, 90, 94]
63	diagnostic probing and active information gathering pressure	[57, 140, 192]
64	world-model disagreement and embodiment pressure	[35, 60, 151]
65	failure recovery and safety-filter pressure	[74, 101, 218]
66	benchmarking, reproducibility, and deployment pressure	[75, 155, 166]
67	uncertainty estimation and calibration pressure	[106, 156, 211]

68	robust control and fallback pressure	[69, 117, 118]
69	diagnostic probing and active information gathering pressure	[12, 40, 203]
70	world-model disagreement and embodiment pressure	[23, 175, 225]
71	failure recovery and safety-filter pressure	[64, 131, 197]
72	benchmarking, reproducibility, and deployment pressure	[16, 121, 145]
73	uncertainty estimation and calibration pressure	[15, 102, 230]
74	robust control and fallback pressure	[34, 37, 212]
75	diagnostic probing and active information gathering pressure	[51, 133, 217]
76	world-model disagreement and embodiment pressure	[138, 173, 215]
77	failure recovery and safety-filter pressure	[196, 202]

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